How LLMs Are Designed to Hallucinate and Reason

Agenda

1) Framing the problem of AI
2) Visualizing the transformer as geometry
3) Case study of addition
4) Reason as linguistic trajectory

There is no artificial intelligence (yet)

Which of these is AI?

(1997) Deep Blue – Defeats Kasparov at chess

(2011) IBM Watson – Wins at Jeopardy!

(2016) AlphaGo/Zero – Wins at Go, Chess, Shogi

(2017) Transformer – LLMs gen. fluent language

(2020) Diffusion Models – Text to image gen.

(2020) Alpha Fold2 – Protein structure unfolding

The transformer as language model was an accident

The Original Goal:
Replace RNNs for Translation

<u>Problems with RNNs:</u>

- Sequential processing (slow)
- Vanishing gradients
- Poor long-range dependency capture

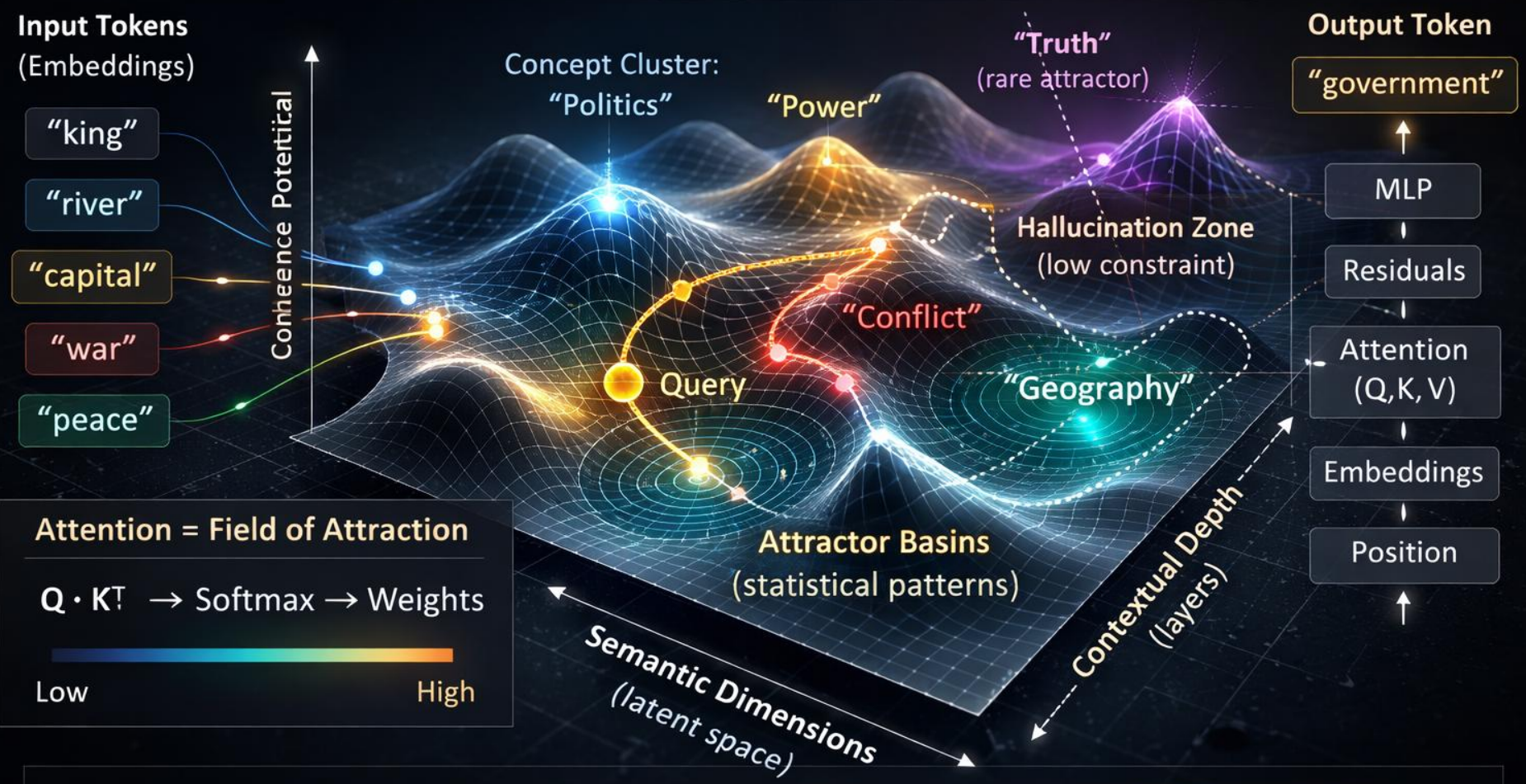
<u>Vaswani's goal:</u>

Build a sequence-to-sequence model that:

- Processes tokens in parallel
- Captures long-range dependencies efficiently
- Is easier to train

The Emergent Semantic Manifold

Attention + Embeddings + Training → Geometric Semantic Field

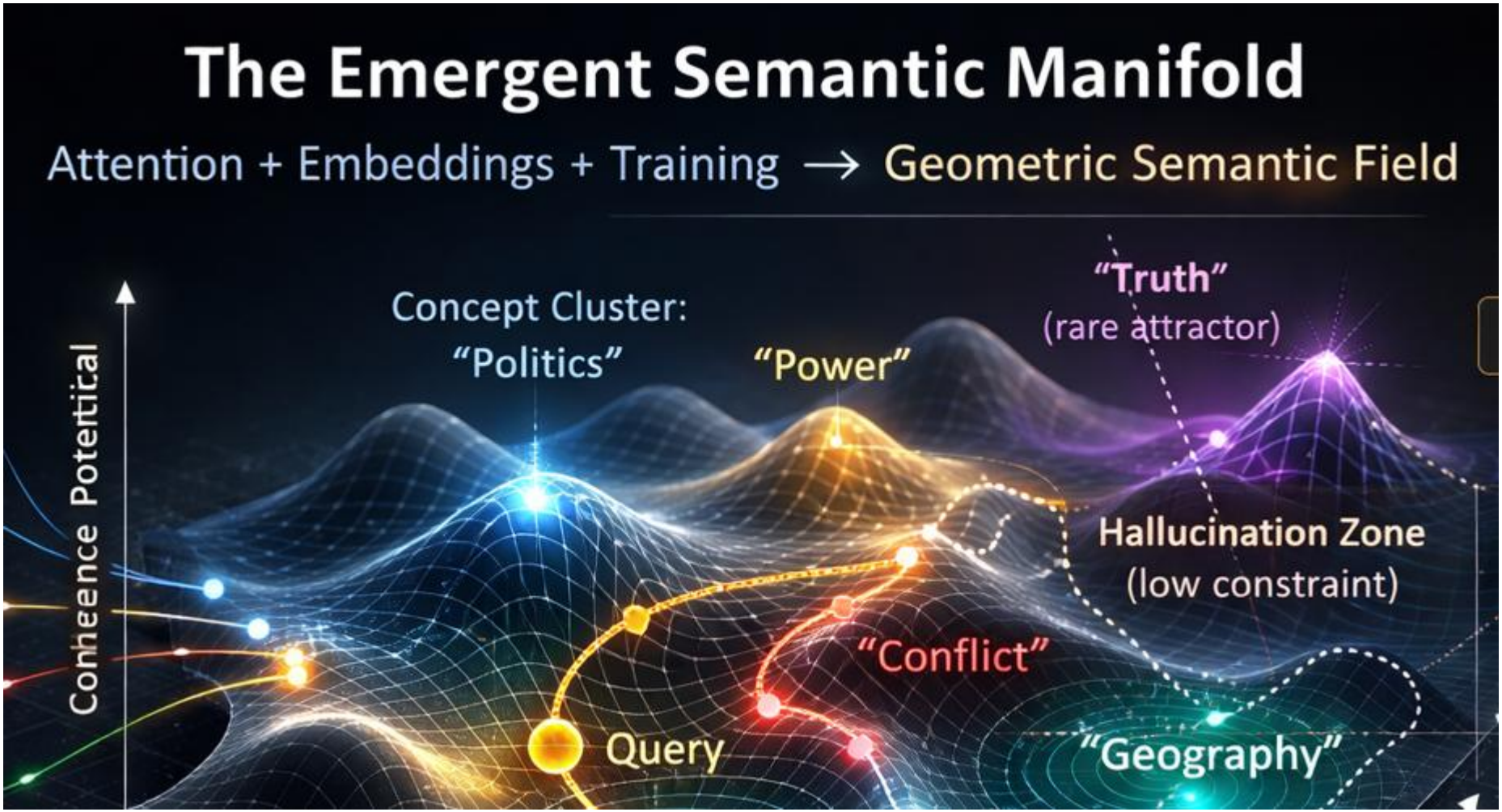


Attention = Field of Attraction

$Q \cdot K^T$ → Softmax → Weights

Low — High

Training (Cross-Entropy) Shapes the Landscape → Not Truth, but Probabilistic Coherence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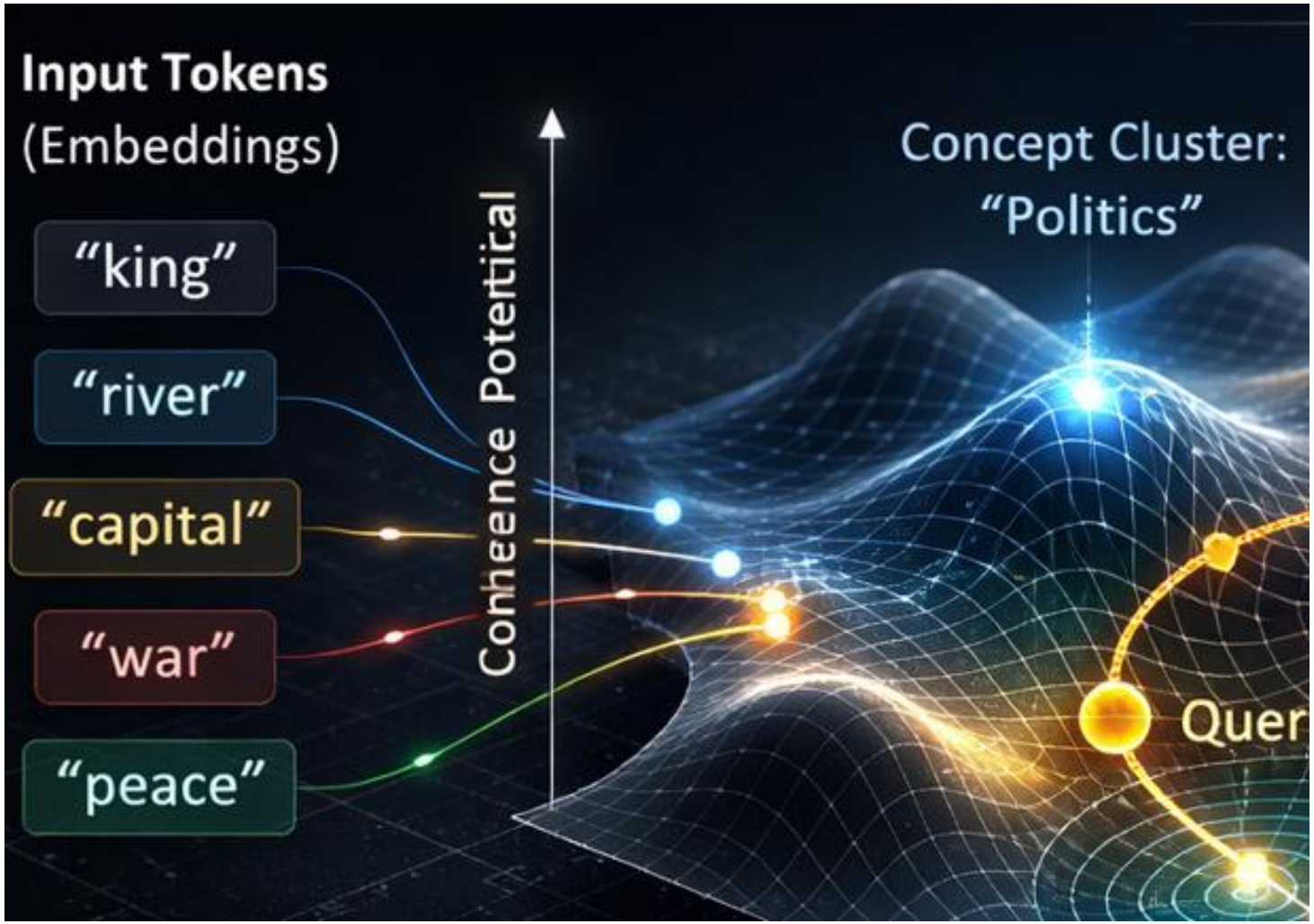
Coheence Potertical
Concept Cluster:
"Politics"
"Power"
"Truth"
(rare attractor)
Hallucination Zone
(low constraint)
"Conflict"
Query
"Geography"


Input Tokens
(Embeddings)
"king"
"river"
"capital"
"war"
"peace"
Coneence Potertical
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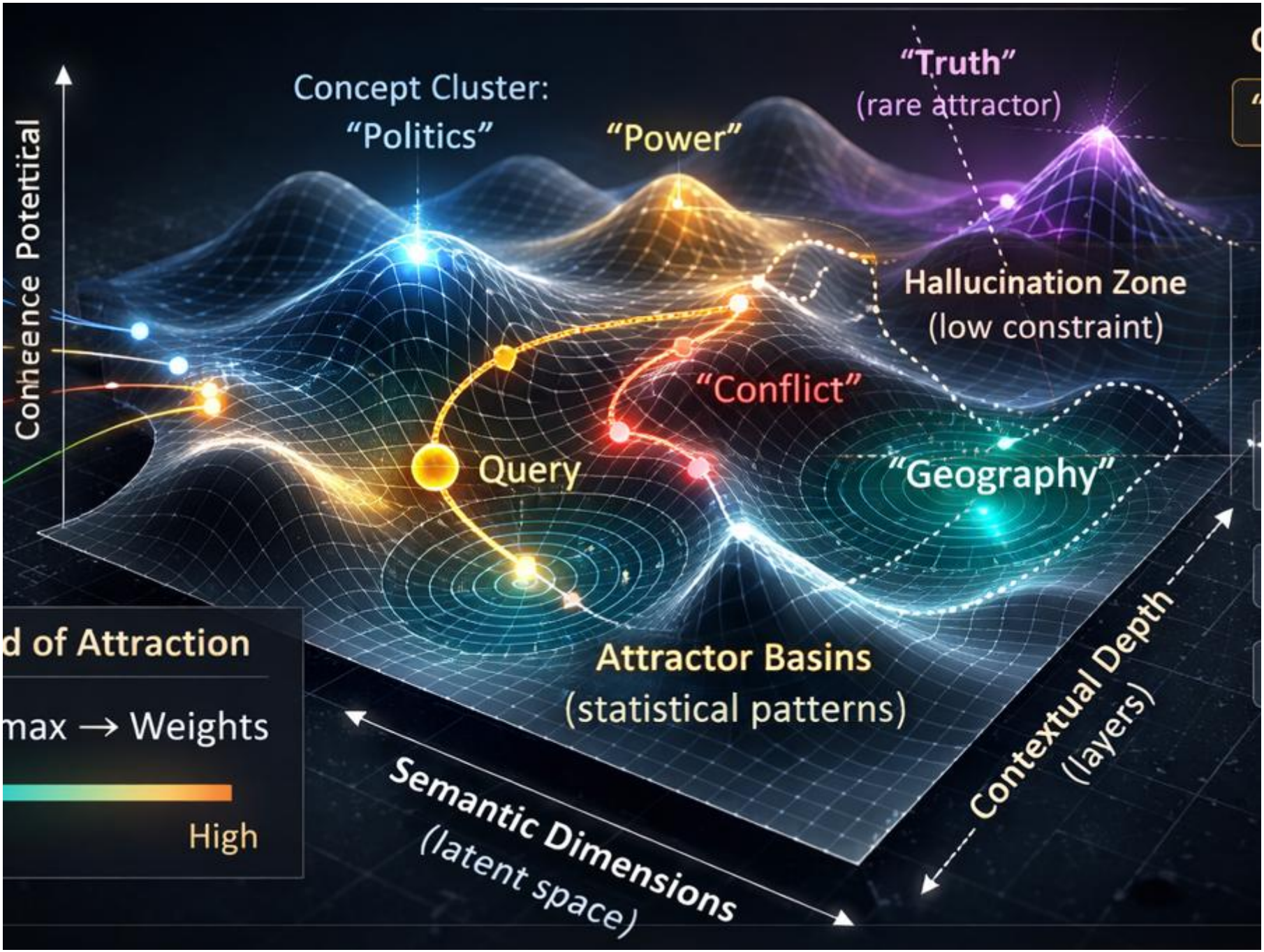
Quer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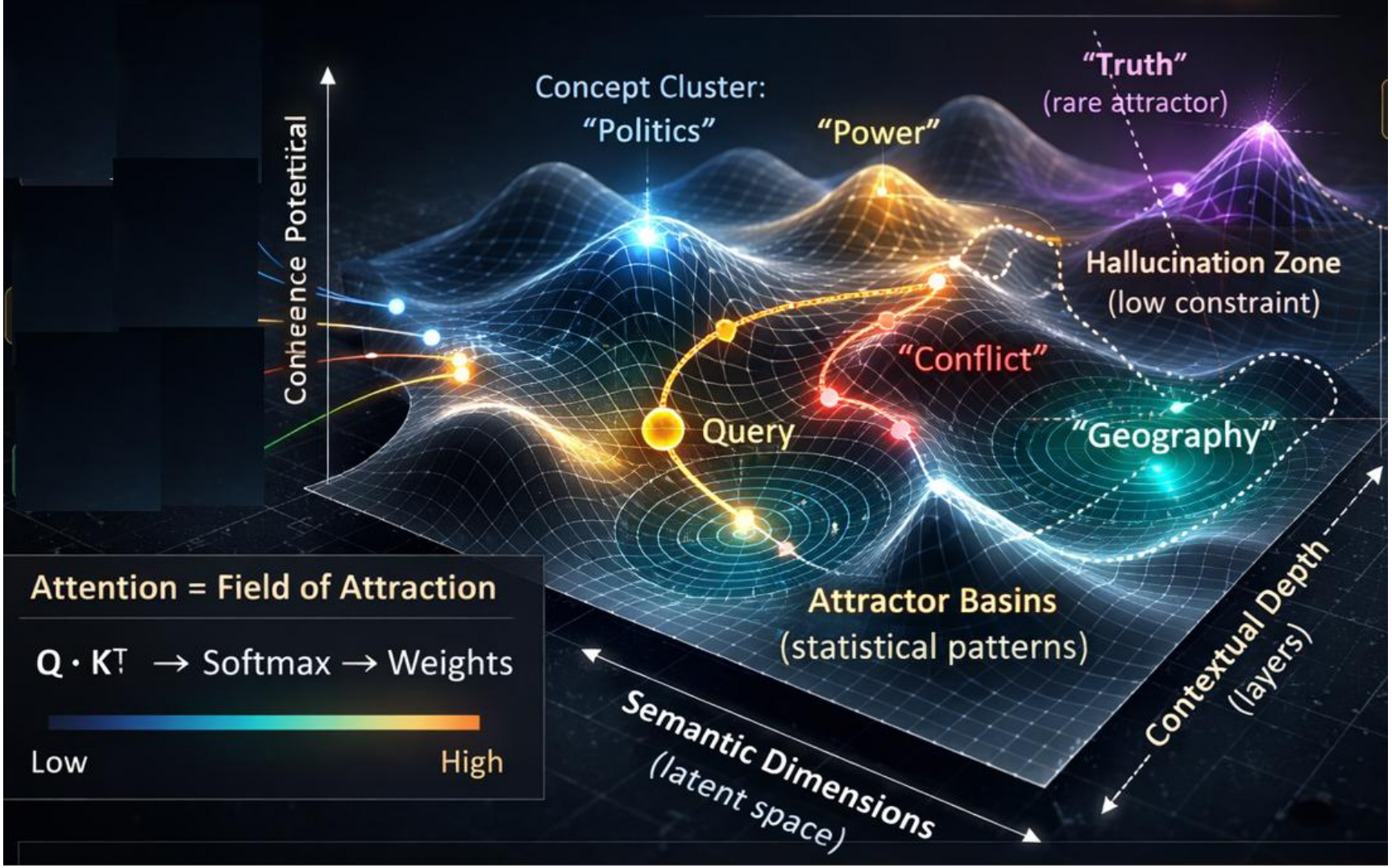
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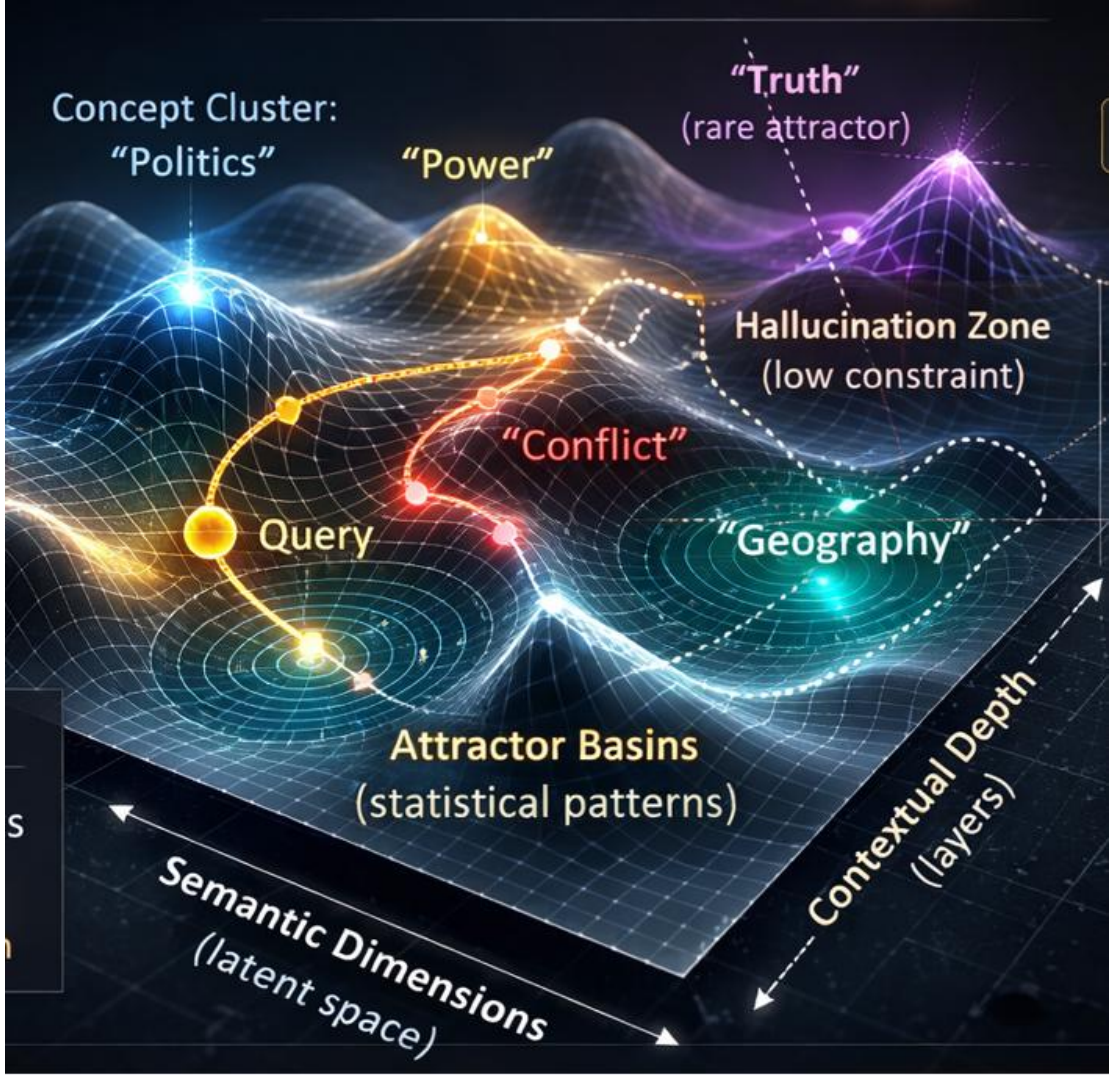
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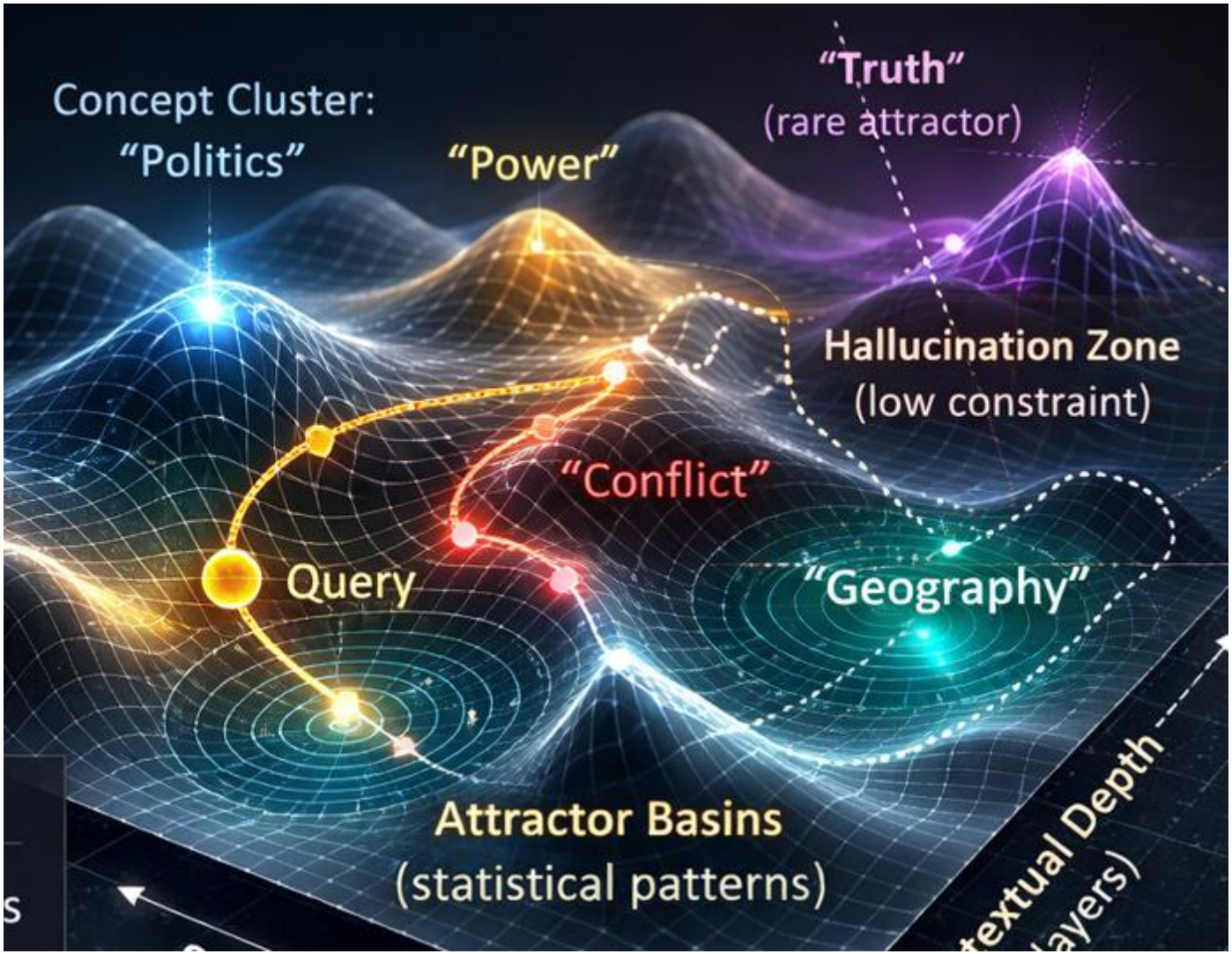
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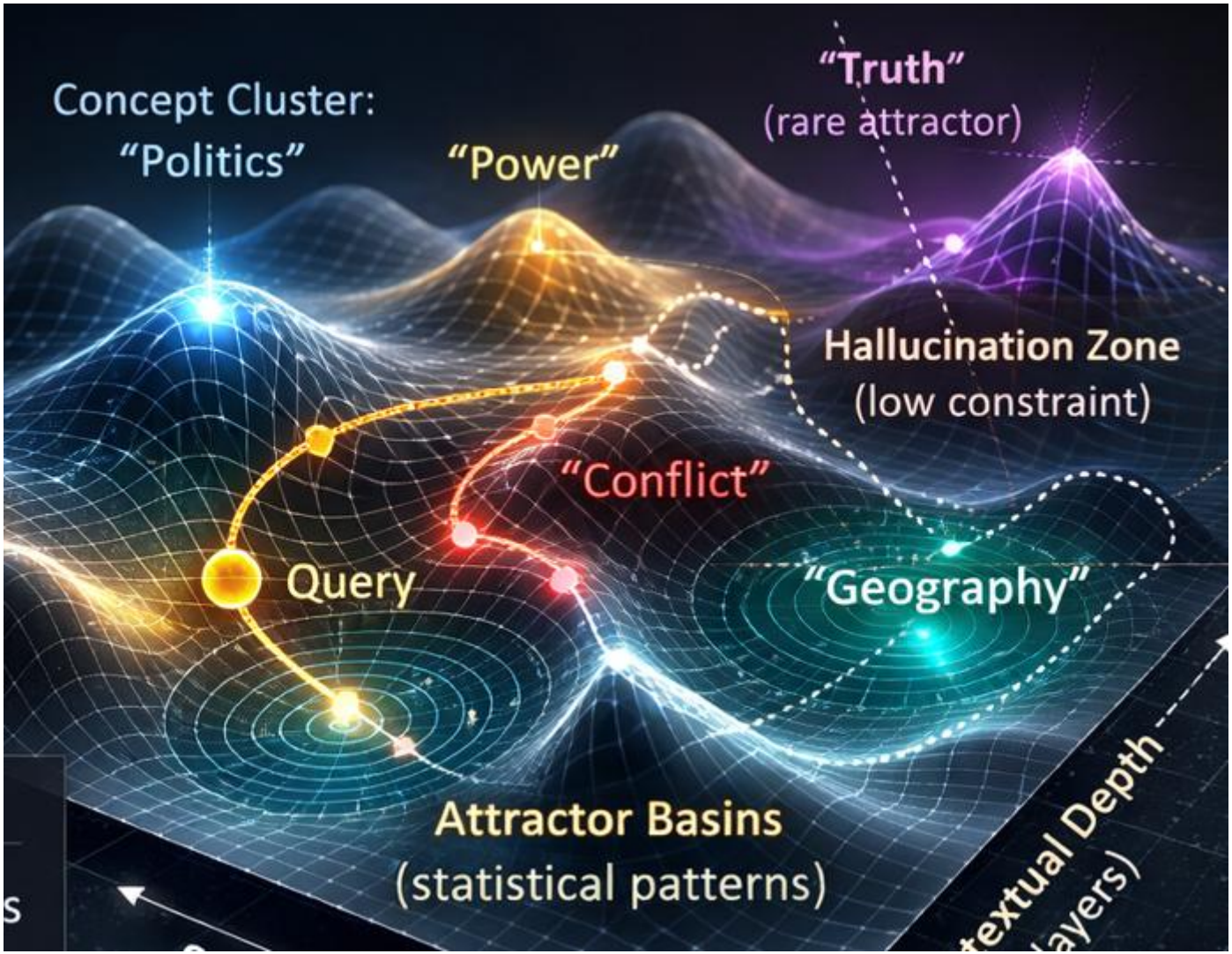
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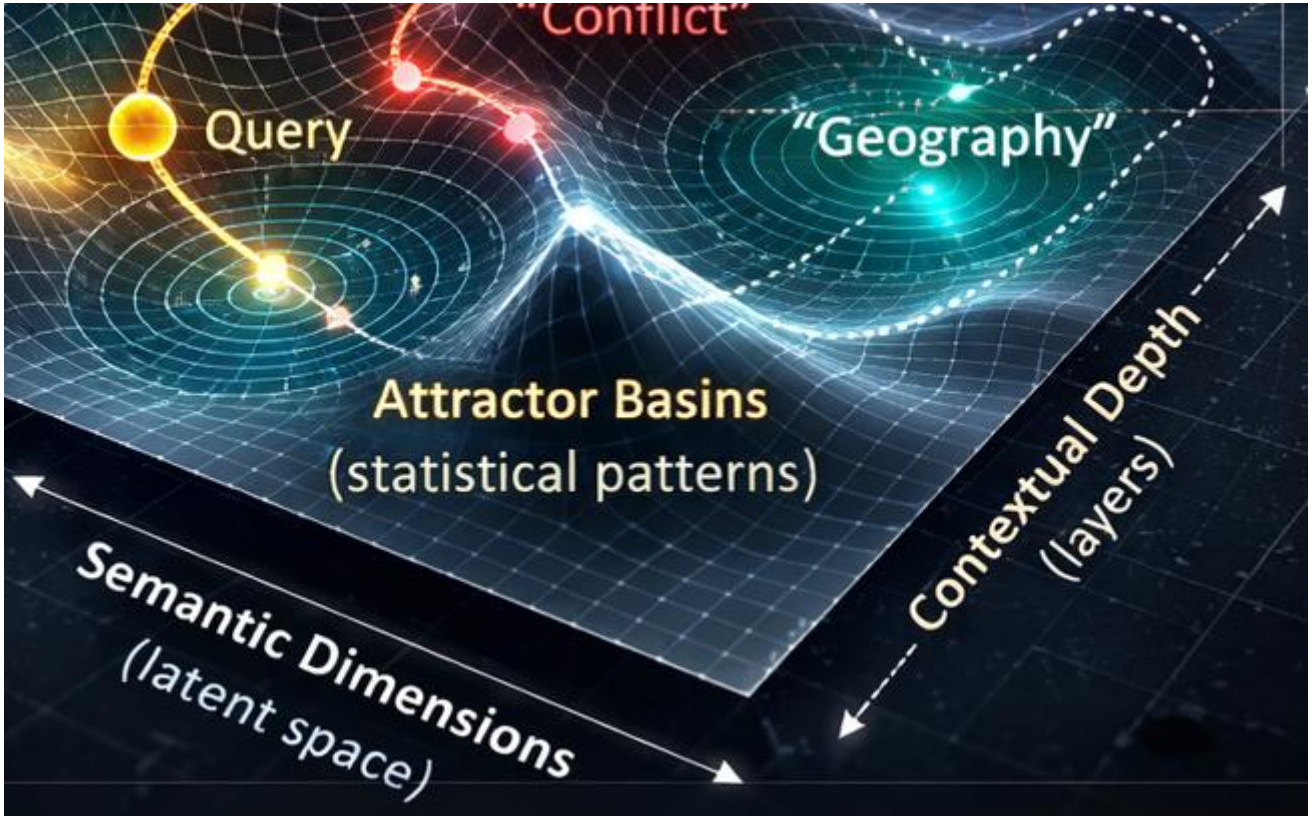
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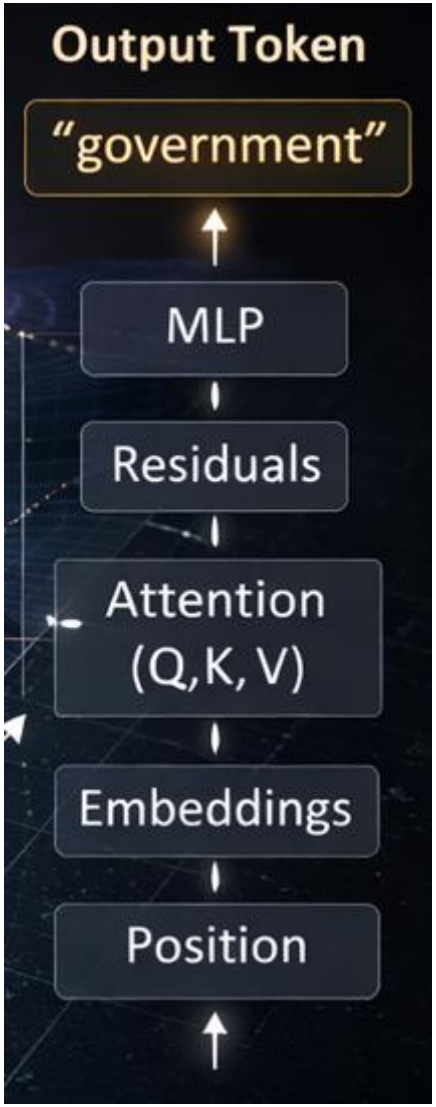
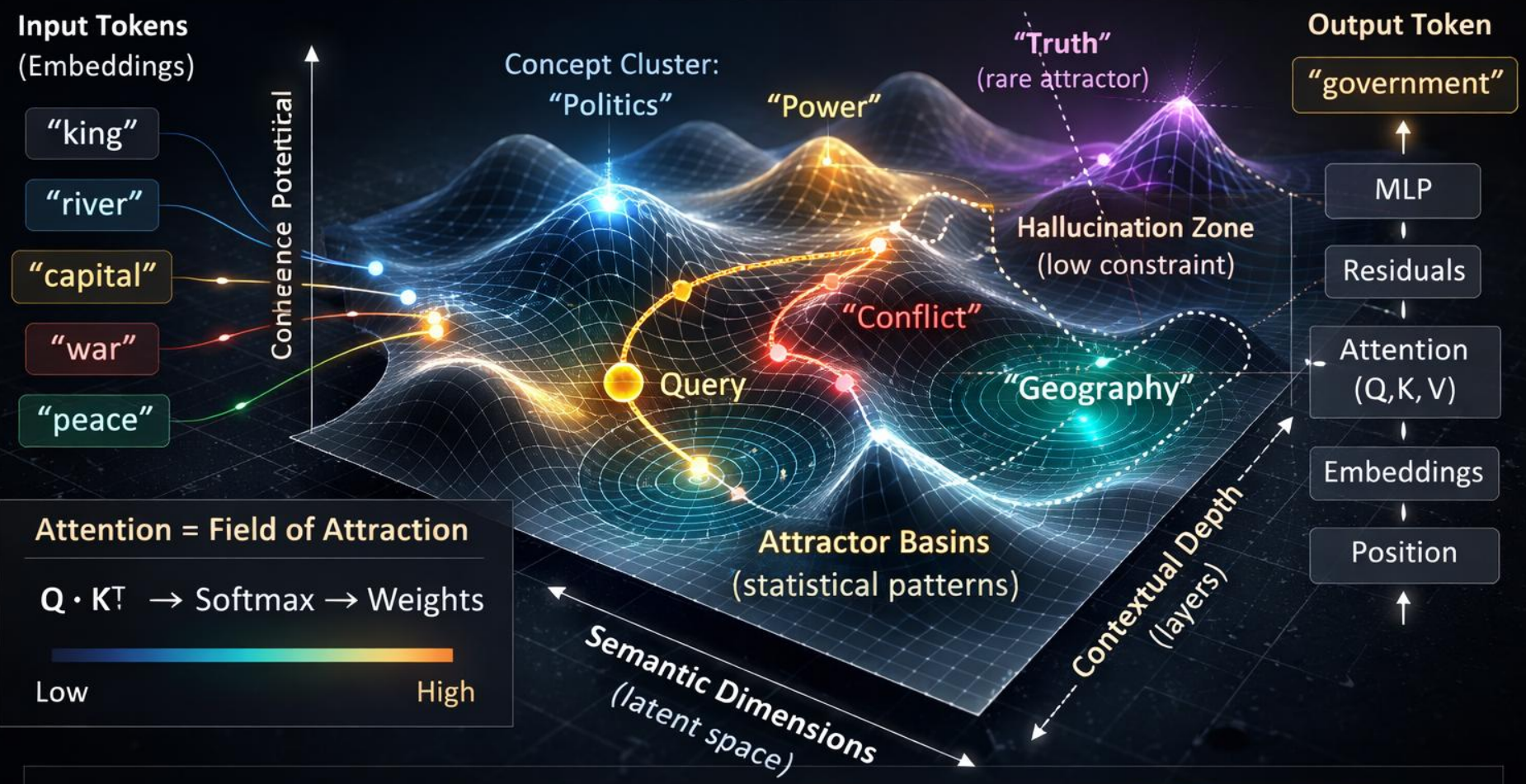

Output Token
"government"
MLP
Residuals
Attention
(Q,K,V)
Embeddings
Position

Training (Cross-Entropy) Shapes the Landscape → Not Truth, but **Probabilistic Coherence**

The Emergent Semantic Manifold

Attention + Embeddings + Training → Geometric Semantic Field



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How is Reasoning Possible in an LLM?

Language Already Contains Structured Thought

Human reasoning is expressed in language through:

- Conditionals ("if…then…")
- Quantifiers ("all," "some")
- Causal markers ("because," "therefore")
- Mathematical notation
- Argument structure
- Stepwise derivations

When the model learns language at massive scale, it is exposed to:

- Proofs
- Explanations
- Debates
- Legal reasoning
- Scientific argumentation
- Mathematical problem solving

How do Humans Add?

Human addition unfolds <u>temporally</u>:

1. Add ones.
2. Carry.
3. Add tens.
4. Resolve.

It has sequential intentional structure.

The transformer's addition:

- Is not temporally unfolded in that way.
- All constraints are active in parallel.
- The "carry" effect is encoded as relational <u>invariants</u>, not as a performed act.

Transformer Case Study: Addition

How does the transformer answer the following query:

$$36 + 59 =$$

Numeric Pattern Features Activate

Now the model begins activating learned arithmetic features.

Interpretability research shows neurons fire for patterns like:

<u>Sum Features</u>

Detect when:

```
a + b ≈ specific value
```

Example:

```
36 + 59 ≈ around 90-100
```

Mod 2, 5, 10 Features

The model also activates features related to:

```
6 + 9 → last digit 5
```

Because:

```
36 → ends in 6
59 → ends in 9
```

Those patterns appear frequently in training data.

So the network predicts:

```
result likely ends in 5
```

Parity Features (Mod 2)

Another invariant:

```
36 even
59 odd
even + odd → odd
```

So, for our example, features pushing toward odd numbers activate.

Constraint Propagation Through Layers

Each transformer layer repeats:

1. Attention
2. Feed-forward (MLP)
3. Residual accumulation

These layers propagate constraints such as:

```
result near 100
result ends in 5
result odd
result compatible with addition
context
```

Each feature contributes a vector direction in representation space.

The network gradually sharpens the prediction.

7. Residual Stream Accumulates Evidence

All features write into a shared **residual stream**.

Think of this like many partial votes:

Feature	Pushes Toward
Ends in 5	95, 85, 75
Magnitude ≈ 100	95, 96, 97
Odd result	95, 97, 99
Addition context arithmetic tokens	

As layers accumulate, **95 becomes the most compatible candidate**.

Logit Projection

The final vector is projected onto the vocabulary.

This produces **logit** scores for every possible token.

Token	Logit
95	12.1
96	10.3
94	9.8
97	9.2

Softmax converts these to probabilities.

Sampling / Decoding

The decoder selects the most likely token:

95

This is emitted as the next token.

Why This Is Not Classical Calculation

Notice what never happened:

- No digit-by-digit carry
- No sequential arithmetic procedure
- No explicit rule application

Instead the model performed:
constraint convergence in vector space.

The correct token wins because it best satisfies the combined statistical structure learned during training.

Why This Discovery Is Important

Before interpretability work, many people thought the model might be:

- memorizing tables
- approximating with heuristics
- doing hidden symbolic reasoning

<u>Instead we now see something different</u>:
The network learns geometric detectors for arithmetic invariants.

Addition emerges from intersecting constraint patterns in vector space.

The Semantic Field as Proto-World

The transformer inhabits a flattened semantic field. Numbers are not quantities. They are relational nodes in a learned pattern manifold.

When prompted with: 36 + 59 =
The model does not “calculate.”
It activates:

- Relations tied to 36
- Relations tied to 59
- Relations tied to "+"
- Relations tied to "="
- Learned invariants about numeric continuation

These features form a temporary structure in activation space.

How is Reasoning Possible in an LLM?

The Missing Piece: Scale

The architecture alone was not enough.

Scale was decisive.

<u>As models grew:</u>

- Parameters increased
- Training data increased
- Context windows increased

<u>The model began to:</u>

- Internalize multi-step patterns
- Approximate algorithms
- Store reusable reasoning templates
- Stabilize long chains of inference

How is Reasoning Possible in an LLM?

The Deep Mechanism
Language is structured relational constraint unfolding.
Reasoning is structured relational constraint unfolding.
They share a structural core.

<u>The transformer models:</u>
High-dimensional relational constraint propagation over sequences.
<u>That is sufficient to:</u>
- Generate fluent language
- Simulate logical inference
- Perform multi-step problem solving
- Approximate algorithmic reasoning

Where the Unity Breaks

However, language modeling and reasoning are not identical.

<u>Language modeling optimizes for:</u>

Statistical continuation.

<u>Reasoning requires:</u>

Normative constraint satisfaction.

<u>The transformer unifies them because:</u>

It treats reasoning as just another kind of text trajectory.

<u>But it does not:</u>

- Evaluate truth
- Check necessity
- Maintain world models
- Enforce non-contradiction
- Bind propositions to reality

Why One Architecture Was Enough

One architecture achieved both because:

- Language encodes reasoning traces.
- Attention encodes relational structure.
- High dimensional embeddings encode abstraction.
- Scale stabilizes multi-step trajectories.
- Softmax forces continuation.

The transformer does not distinguish:

Between poetic metaphor and logical deduction.
Both are trajectories through semantic space.